How Artificial Intelligence Research Has Spread Across the Globe

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***Abstract*—Research is a constantly evolving process that requires a great deal of understanding even for the most simple concepts. This understanding is often greatly hindered by an overfocus on the purely theoretical and practical aspects of the topics in question. Although such aspects are important, many people exclude other contributing factors, such as the history of a given topic. In this text, we seek to provide some small insight into how research spreads and changes over time, using the OpenAlex [1] database. Specifically, we opted to focus on artificial intelligence (AI), trying to gain insight into how it spreads both geographically and semantically, though we ended up only being able to look into the geographic spread in the end.**

***Index Terms*—Artificial Intelligence, Machine Learning, Topic Spread, Datamining, OpenAlex, Geographic Spread**

I. Introduction

The purpose of this project was to apply the knowledge we had learned about data mining throughout the semester and to generate a practical report on our findings.

We were given access to a Virtual Machine (VM) that contains a snapshot of the OpenAlex [1] database from May 2025. Additionally, this VM is equipped with python and various tools such as pandas [2], mlxtend, and various others.

We were given free reign to research whatever we wished. We ended up choosing machine learning (ML) as the subject of our research for a variety of reasons. The most obvious reason being how much artificial intelligence (AI) has grown in just the past five years alone. We both have an interest in machine learning and figured: "What better way is there to learn more about it than to research how it spread?"

With this fundamental question in mind, we began to refine it into a couple of more detailed questions, one focused on what regions of the world have produced related research (Section III) and how the concepts related to artificial intelligence have merged and diverged (Semantic Spread). Unfortunately, we were unable to complete the semantic spread portion of our research in a timely manner, but we believe it would be a good direction to go in the future.

II. Tools and Setup

We were given several choices on how we should collect and manipulate our data, including, of course, the pandas python package. Most of our fellow students will have opted to primarily use this library for data intake, but we chose instead to use DuckDB [3], an open-source in-memory database solution which allows us to directly run SQL queries on our data. As the data is stored in both CSV and Parquet formats, we thought creating tables and views referencing the files in question would allow for a smoother more consistent querying process. We were correct, to a point, but we did end up running into some issues, to be discussed later. Much like pandas, DuckDB is capable of lazy-loading, a process where data is only loaded into memory when it is needed, allowing the majority of the processing power and memory of the VM to be used elsewhere.

III. Geographic Spread

A. Question

Our research into geographic spread started with a single question: "How much of a role does physical proximity play in the spread of topics, concepts, and subfields related to artificial intelligence in research papers?"

B. Hypothesis

We figured that the extremely common presence of the Internet and various other networks would significantly reduce the impact of physical location on topic spread. Research is fundamentally information-based, and information networks that allow transmission at nearly the speed of light are far more efficient than needing to travel to a location to see research. We did take into account the massive computational requirements of modern machine learning, which increases the odds of physical constraints, as larger server banks owned by specific entities are not likely to be used by unaffiliated researchers, and there is a high cost to their construction.

C. Initial Approach and Results

Our OpenAlex snapshot is several terabytes in size, a scale which makes data handling very difficult. Instead of looking into every possible related topic over the the entirety of the dataset, we chose to limit our timeframe to be between the years of 2010 and 2019 and we chose only ten topics to focus on:

- Face and Expression Recognition
- Data Mining Algorithms and Applications
- Quantum Computing Algorithms and Architecture

- Image Retrieval and Classification Techniques
- Adversarial Robustness in Machine Learning
- Multimodal Machine Learning Applications
- Machine Learning and Algorithms
- Machine Learning and Data Classification
- Machine Learning and ELM
- Machine Learning in Healthcare

These topics acted as markers in a given paper, and along with information about the papers' publishing institutions, and the location of those institutions, we used the k-means clustering algorithm with a haversine distance function to create a few initial clusters. We then checked their silhouette scores to refine our clustering. We discovered that the optimal number of hubs was approximately three topic, in order to get a global distribution. The top three clusters for 2010 can be seen in Figure 1.

D. Updated Approach

With these initial findings, we opted to run an HDBSCAN algorithm to see if anything would change.

This run ended up giving us 168 individual hubs and 783 institutions classified as noise. This new set of clusters had a silhouette score of 0.6903. Once we were satisfied with our clustering, we used both clustering methods across all of the years in our timeframe, giving us k-means cluster for 2015 in Figure 2, and for 2019 in Figure 3. The second clustering method was used to calculate weighted centers for each topic across the years. These are shown in Figures 4, 5, and 6.

E. Updated Results

Four of the ten topics directly supported our hypothesis:

- Adversarial Robustness in Machine Learning
- Multimodal Machine Learning Applications
- Machine Learning and Data Classification
- Machine Learning in Healthcare

The former two saw increased growth across the decade compared to the others (Figures 7 and 8), and the latter two had large spikes in the weighted center's motion across the years (Figures 9 and 10).

There were two topics which contradicted our hypothesis, both of which had relatively small changes in their weighted centers (Figures 11 and 12):

- Quantum Computing Algorithms and Architecture
- Machine Learning and ELM

Both of these topics require highly specialized equipment, so it makes sense that research into them tends to stay relatively close to where the equipment already is. We found that the increase in concentration of research outpaced the spread of the topic, so while they did spread, they tend towards depth over breadth.

Interestingly, both *Data Mining Algorithms and Applications* and *Image Retrieval and Classification Techniques* saw decreases in research (Figures 13 and 14). These topics seemed to be spreading at first glance, but it is more accurate to say that they moved, as the hubs all settled in China by the end of the decade, which would appear to be an increase if we didn't also have the counts of the research papers including them.

Finally, we also had two topics that remained fairly stable:

- Face and Expression Recognition
  Figure 15
- Machine Learning and Algorithms
  Figure 16

It is possible that choosing one decade to look at did not properly capture the growth of these two topics, but from this decade alone they seemed very stable.

F. Conclusion

We believe that artificial intelligence is so broad a subject that, although our hypothesis makes sense, there is enough chaos that the various topics need to be looked at on a case-by-case basis, and singular judgment is not currently possible. We may have come to a different conclusion with a larger sample size, or using different topics, but our current data both supports and contradicts our hypothesis. There is no definitive evidence for one side or the other.

It is possible that this is because topics that appear similar may actually vary greatly, but it may also be the case that the various topics are in a state of unstable equilibrium, where they are still trying to stabilize.

IV. Future Directions

A. Learning Experience

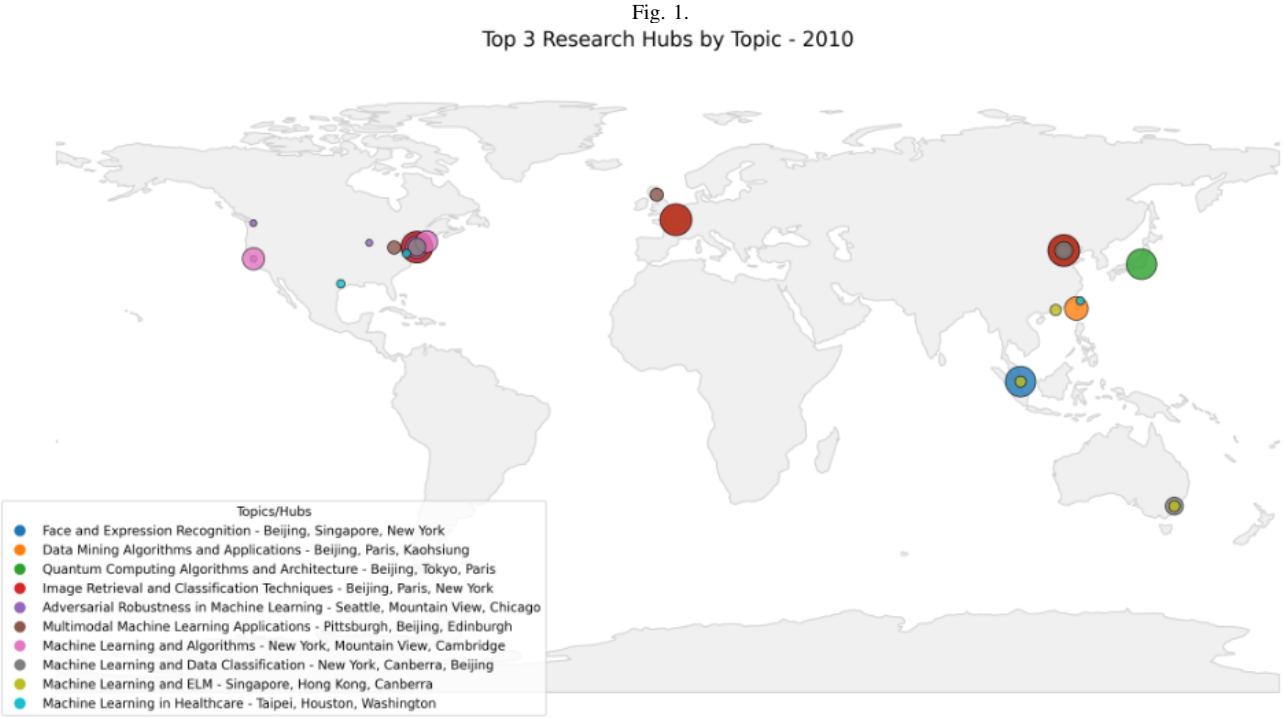
We believe that our research was too broad, even only looking at Machine learning. We needed a minimum of a decade for the temporal factor of our research, but perhaps we should have narrowed our topics even more, or found topics related to a subtopic instead of the overly broad subject of machine learning. We could also have used a dynamic time component instead of using just a singular decade across all of them; we could have looked further into when the individual topics emerged.

We also found that using DuckDB was harder than expected. It was more flexible and capable than pandas, sure, but it used a lot more memory, crashed the kernel often, and required some serious fine tuning to get right. If we could go back, we probably would have either chosen pandas to work with, or we would have spent more time fine tuning our connections and models before going deeper into the database.

Issues with DuckDB are precisely the reason we did not complete the semantic spread portion of our research. We simply did not have enough time to make headway by the time we got it working in the way we needed to.

Along with learning about the subject and the tools, we learned more about research projects. This particular project was far more demanding than what we had experienced in the past, and we were not nearly as prepared as we should have been. We experiences both scope creep and capability overestimation, which compounded on each other and caused serious issues, especially around keeping track of our code and making sure it was understandable and fully functional. Small issues like not deleting a single variable after it was

Fig. 1.



done would result in major time loss, but be incredibly hard to detect once in place, because we had very little feedback from the program itself, even if we added logging.

B. Future Research Paths

Although we would have liked to do more with this project, we were significantly limited. As such, we have come up with two potential future research project ideas which should build off of our existing research.

The first idea is, of course, semantic spread. If we can Identify how topics actually emerge as fields of study, from their keywords and other related topics, we may be able to better predict what new research paths will emerge in the future. Telling the future outright is impossible, but identifying new topics with potential is a highly useful capability, and using techniques from natural language processing seems like an effective way to achieve it.

The second idea is identifying how topics move together. Instead of focusing specifically on how a single topic spreads, it may be insightful to identify how seemingly completely unlinked topics actually share patterns. If a given machine learning topic is somehow related to a specific medical topic, when they don't have any common-sense correlation, we may be able to find connections which increase our knowledge of both.

References

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[3] H. Mühleisen and M. Raasveldt, *Duckdb: Dbi package for the duckdb database management system*, R package version 1.5.2, 2026. [Online]. Available: https://r.duckdb.org/.

Fig. 2.

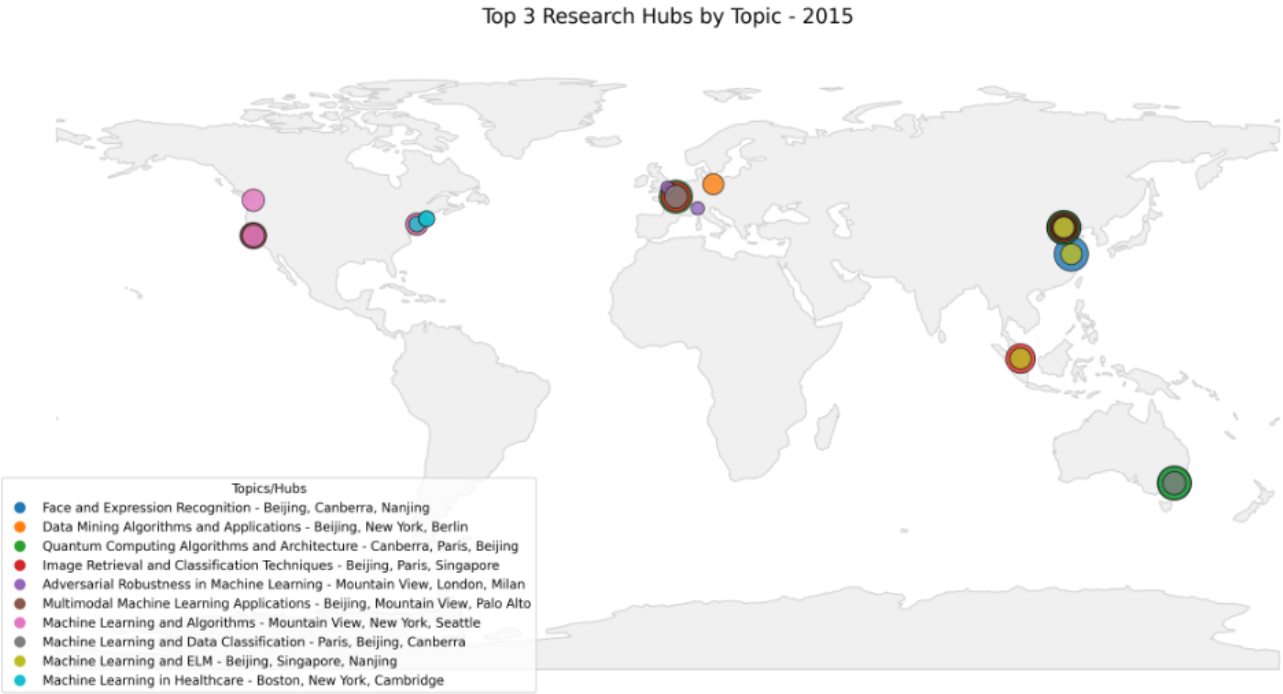

Top 3 Research Hubs by Topic - 2015
Topics/Hubs
Face and Expression Recognition - Beijing, Canberra, Nanjing
Data Mining Algorithms and Applications - Beijing, New York, Berlin
Quantum Computing Algorithms and Architecture - Canberra, Paris, Beijing
Image Retrieval and Classification Techniques - Beijing, Paris, Singapore
Adversarial Robustness in Machine Learning - Mountain View, London, Milan
Multimodal Machine Learning Applications - Beijing, Mountain View, Palo Alto
Machine Learning and Algorithms - Mountain View, New York, Seattle
Machine Learning and Data Classification - Paris, Beijing, Canberra
Machine Learning and ELM - Beijing, Singapore, Nanjing
Machine Learning in Healthcare - Boston, New York, Cambridge

Fig. 3.

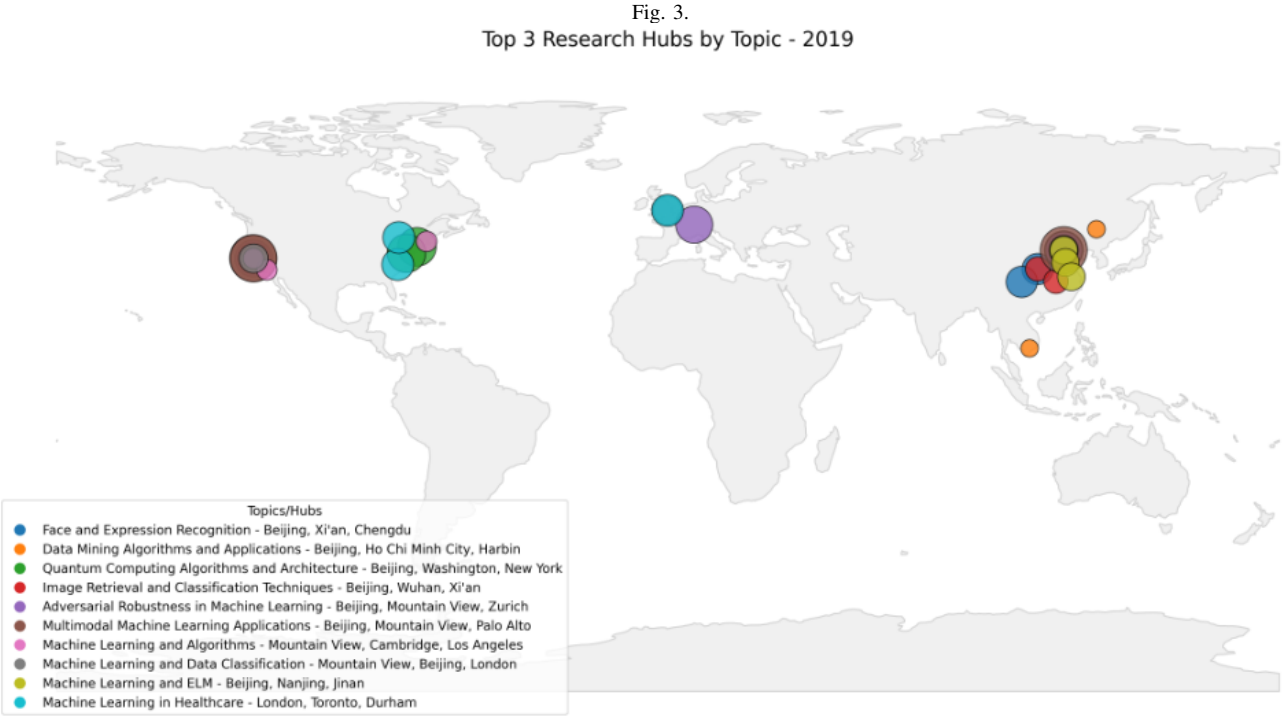

Top 3 Research Hubs by Topic - 2019
Topics/Hubs
Face and Expression Recognition - Beijing, Xi'an, Chengdu
Data Mining Algorithms and Applications - Beijing, Ho Chi Minh City, Harbin
Quantum Computing Algorithms and Architecture - Beijing, Washington, New York
Image Retrieval and Classification Techniques - Beijing, Wuhan, Xi'an
Adversarial Robustness in Machine Learning - Beijing, Mountain View, Zurich
Multimodal Machine Learning Applications - Beijing, Mountain View, Palo Alto
Machine Learning and Algorithms - Mountain View, Cambridge, Los Angeles
Machine Learning and Data Classification - Mountain View, Beijing, London
Machine Learning and ELM - Beijing, Nanjing, Jinan
Machine Learning in Healthcare - London, Toronto, Durham

Fig. 4.
Topic weighted centers - 2010

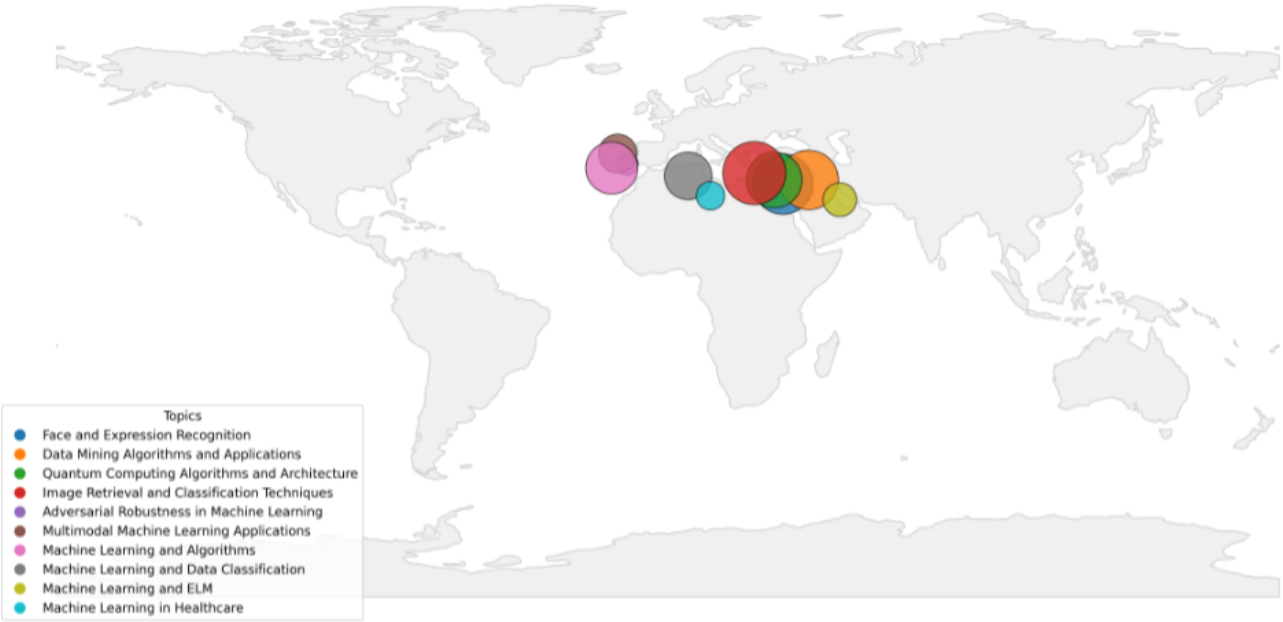


Fig. 5.
Topic weighted centers - 2015

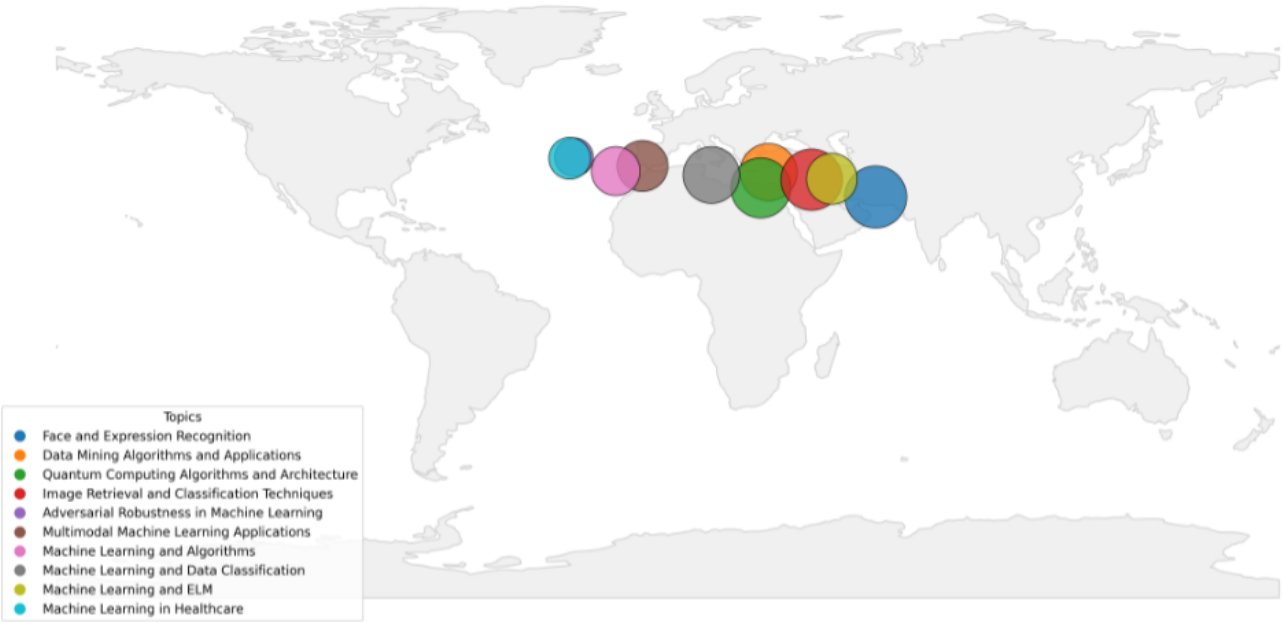


Fig. 6.

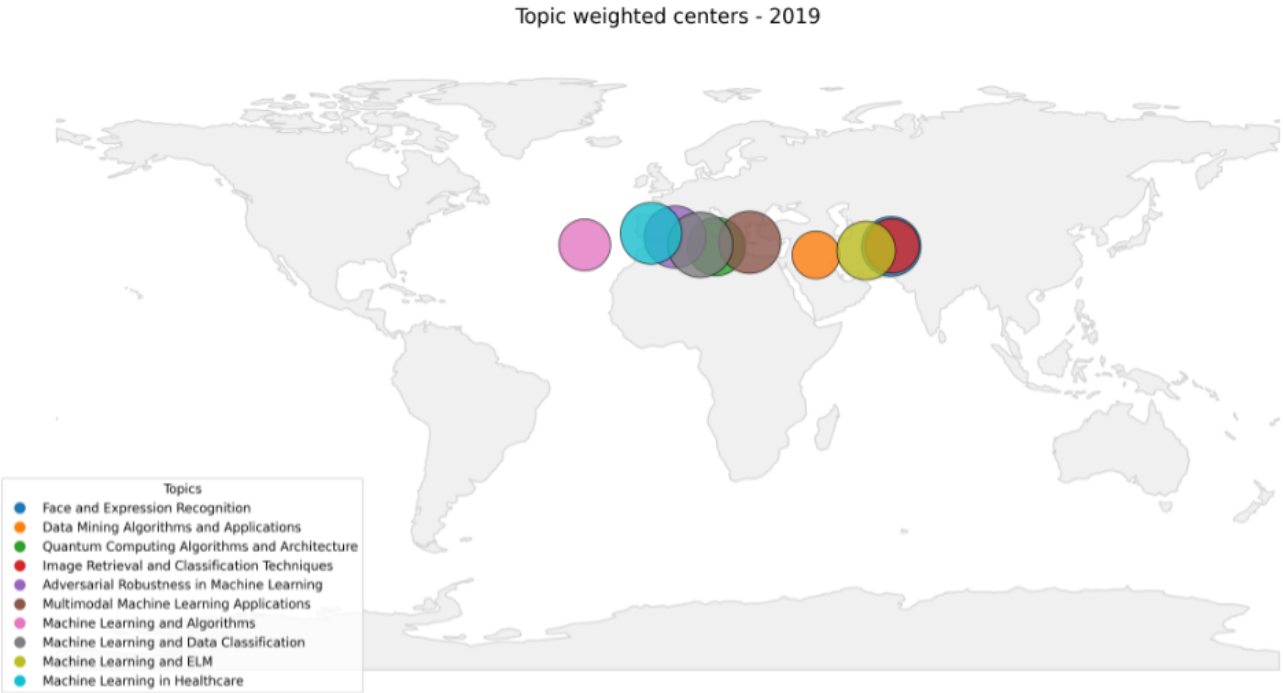
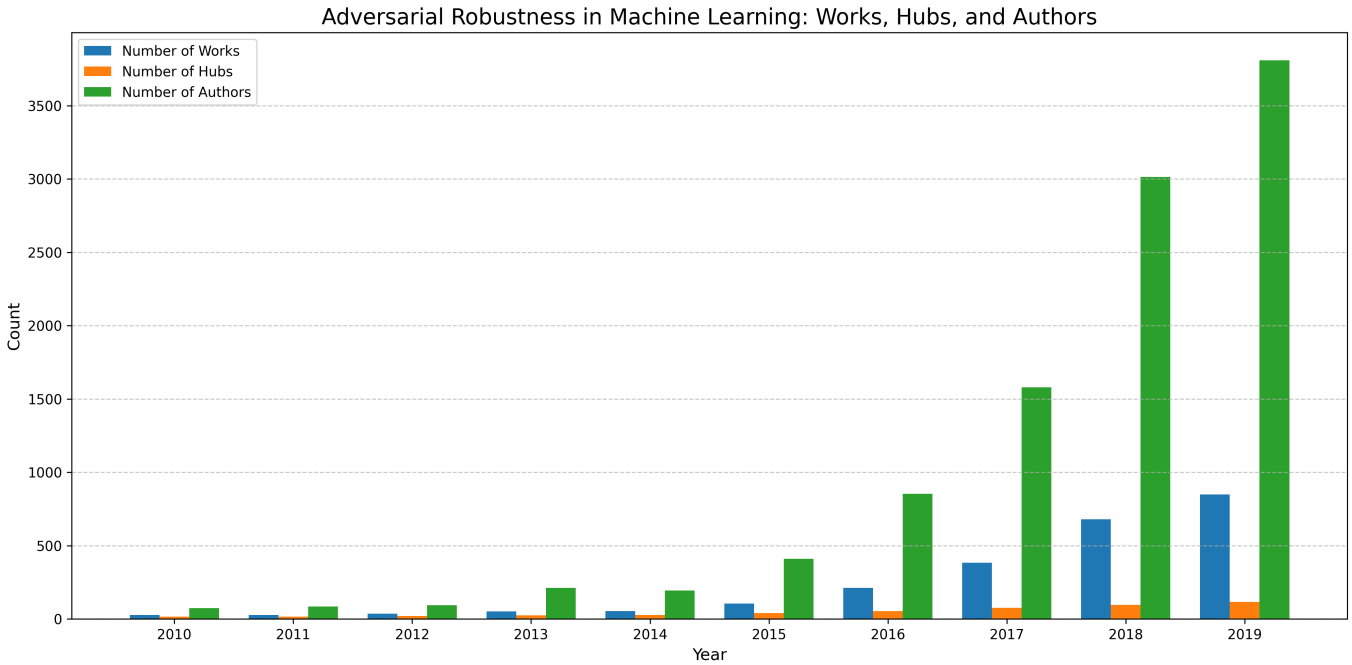

Topic weighted centers - 2019
Topics
Face and Expression Recognition
Data Mining Algorithms and Applications
Quantum Computing Algorithms and Architecture
Image Retrieval and Classification Techniques
Adversarial Robustness in Machine Learning
Multimodal Machine Learning Applications
Machine Learning and Algorithms
Machine Learning and Data Classification
Machine Learning and ELM
Machine Learning in Healthcare

Fig. 7.


Adversarial Robustness in Machine Learning: Works, Hubs, and Authors
Number of Works
Number of Hubs
Number of Authors
Count
0
500
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1500
2000
2500
3000
3500
2010
2011
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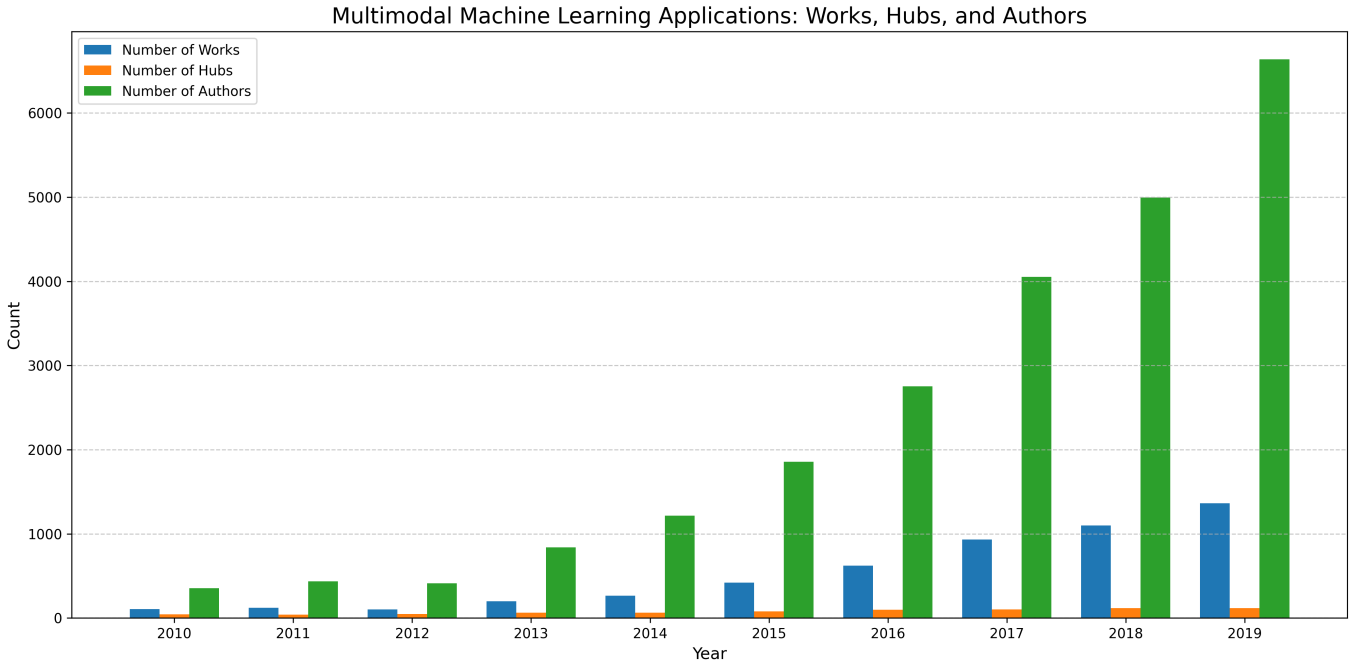
Year

Fig. 8.


Multimodal Machine Learning Applications: Works, Hubs, and Authors
Number of Works
Number of Hubs
Number of Authors
Count
0
1000
2000
3000
4000
5000
6000
2010
2011
2012
2013
2014
2015
2016
2017
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2019
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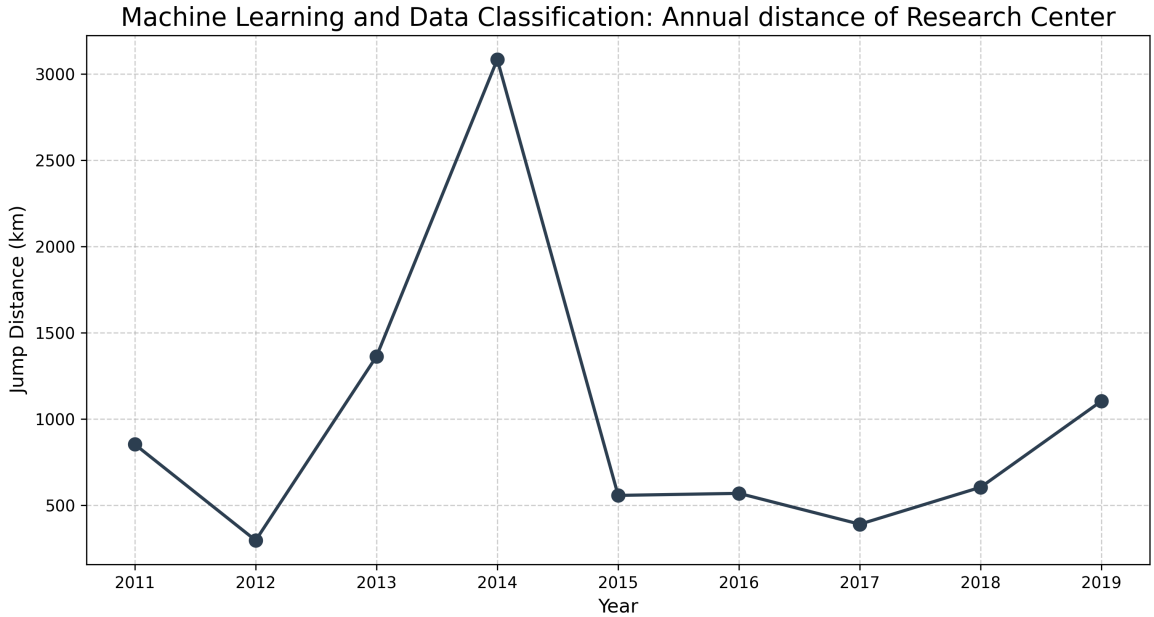
Fig. 9.


Machine Learning and Data Classification: Annual distance of Research Center
Jump Distance (km)
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2011
2012
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2018
2019
Year

Fig. 10.

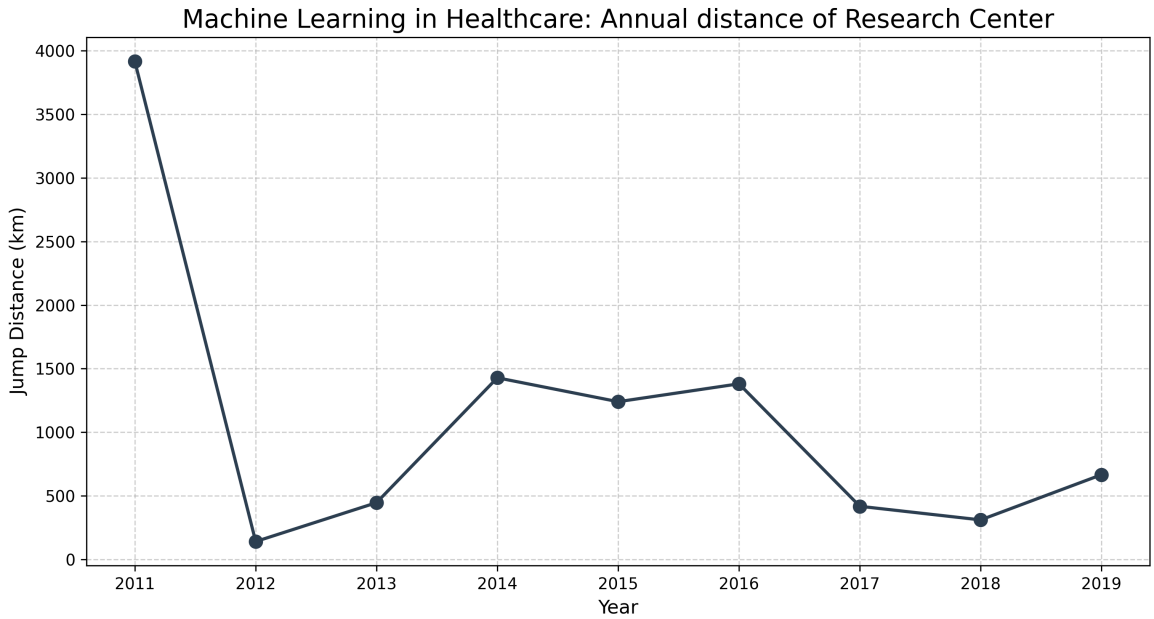
Machine Learning in Healthcare: Annual distance of Research Center
Jump Distance (km)
4000
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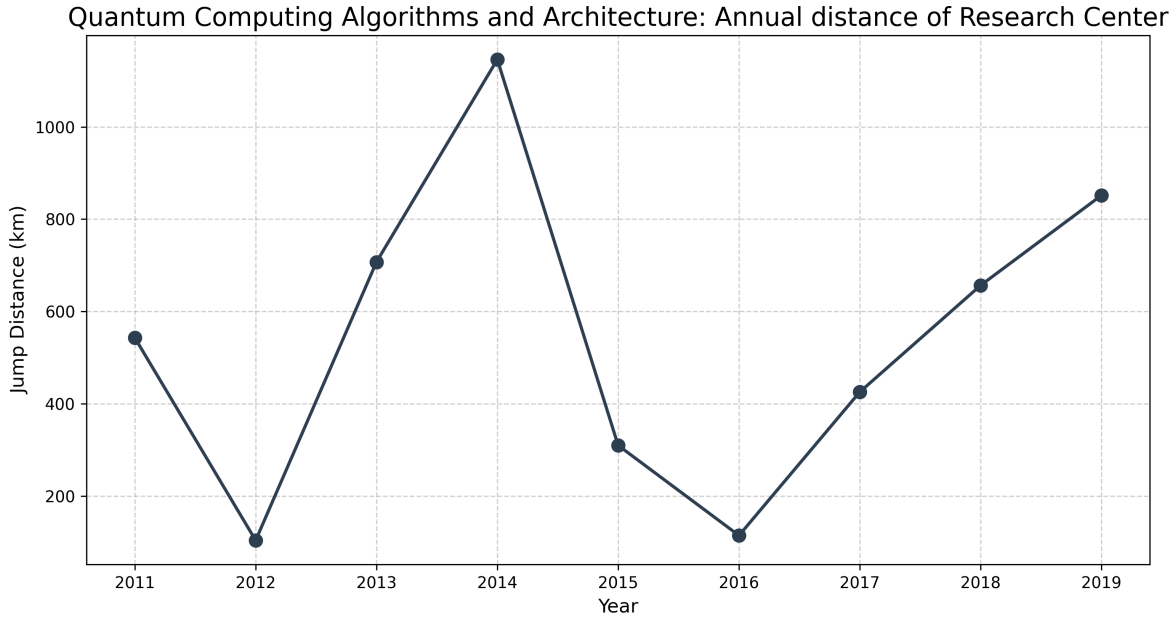
Fig. 11.


Quantum Computing Algorithms and Architecture: Annual distance of Research Center
Jump Distance (km)
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2019
Year

Fig. 12.

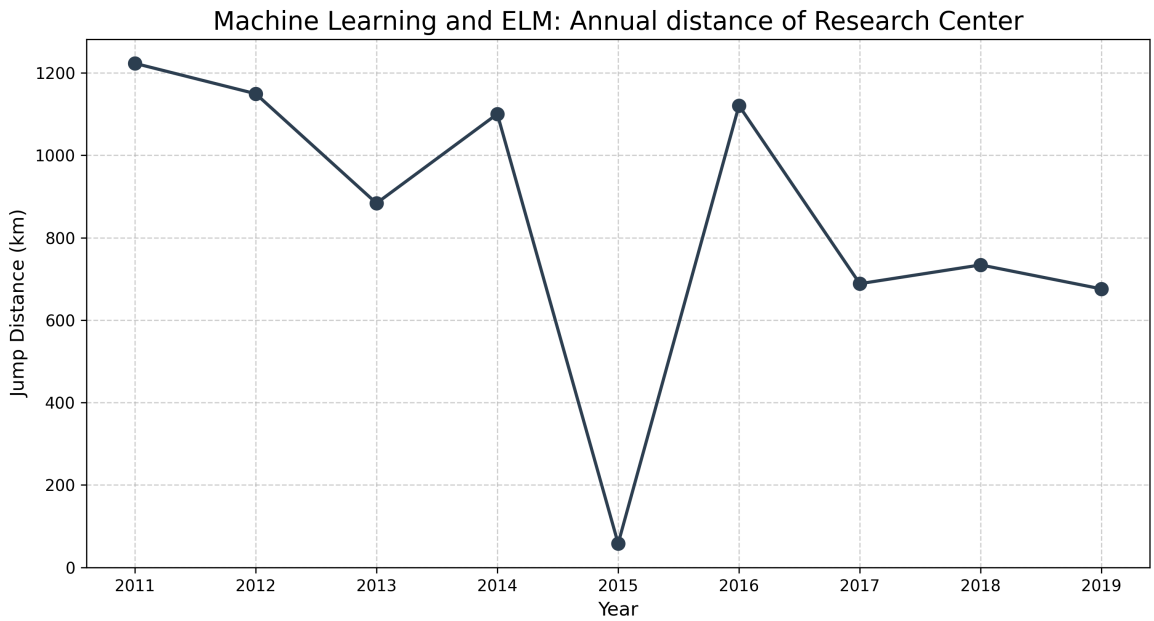
Machine Learning and ELM: Annual distance of Research Center
Jump Distance (km)
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Fig. 13.

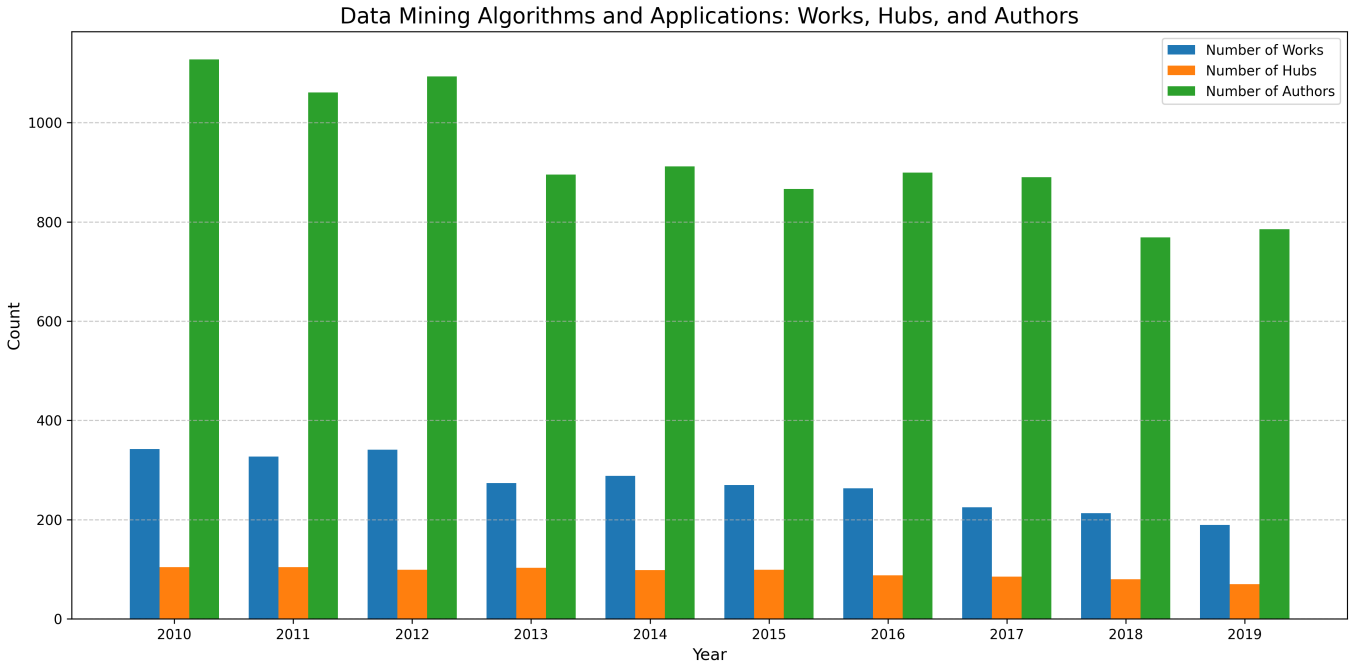
Data Mining Algorithms and Applications: Works, Hubs, and Authors
Number of Works
Number of Hubs
Number of Authors
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Fig. 14.

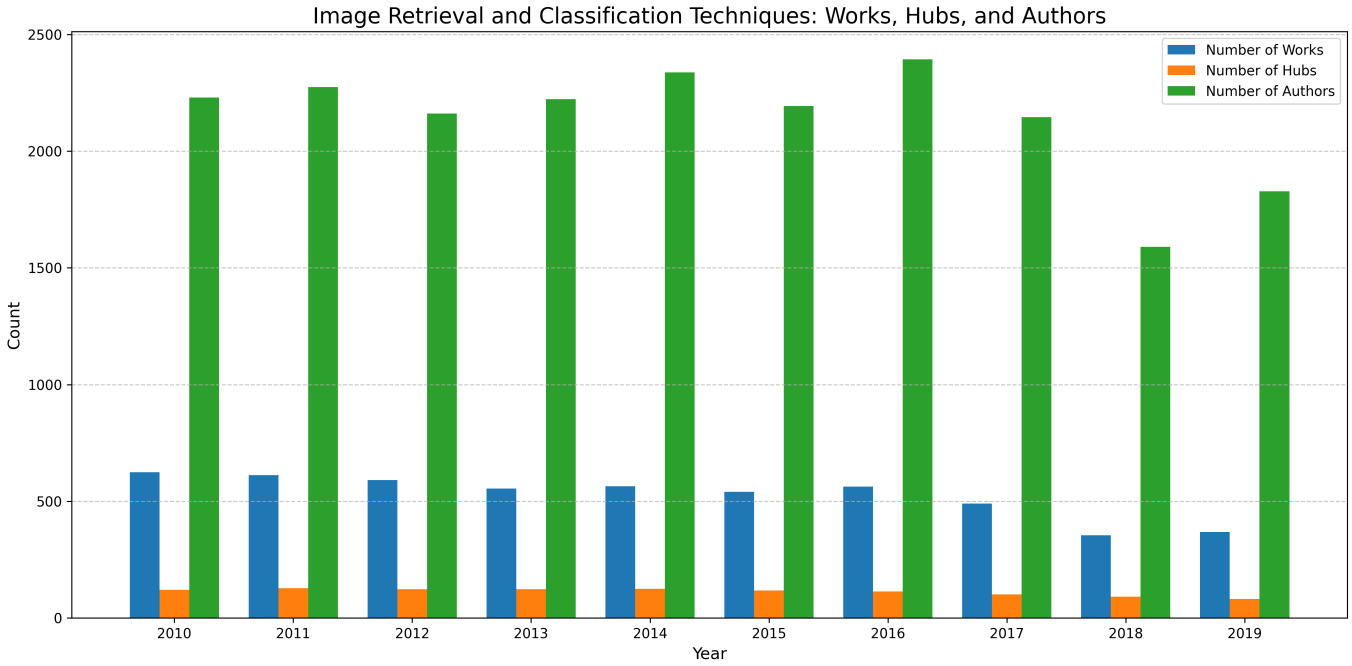
Image Retrieval and Classification Techniques: Works, Hubs, and Authors
Number of Works
Number of Hubs
Number of Authors
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Fig. 15.

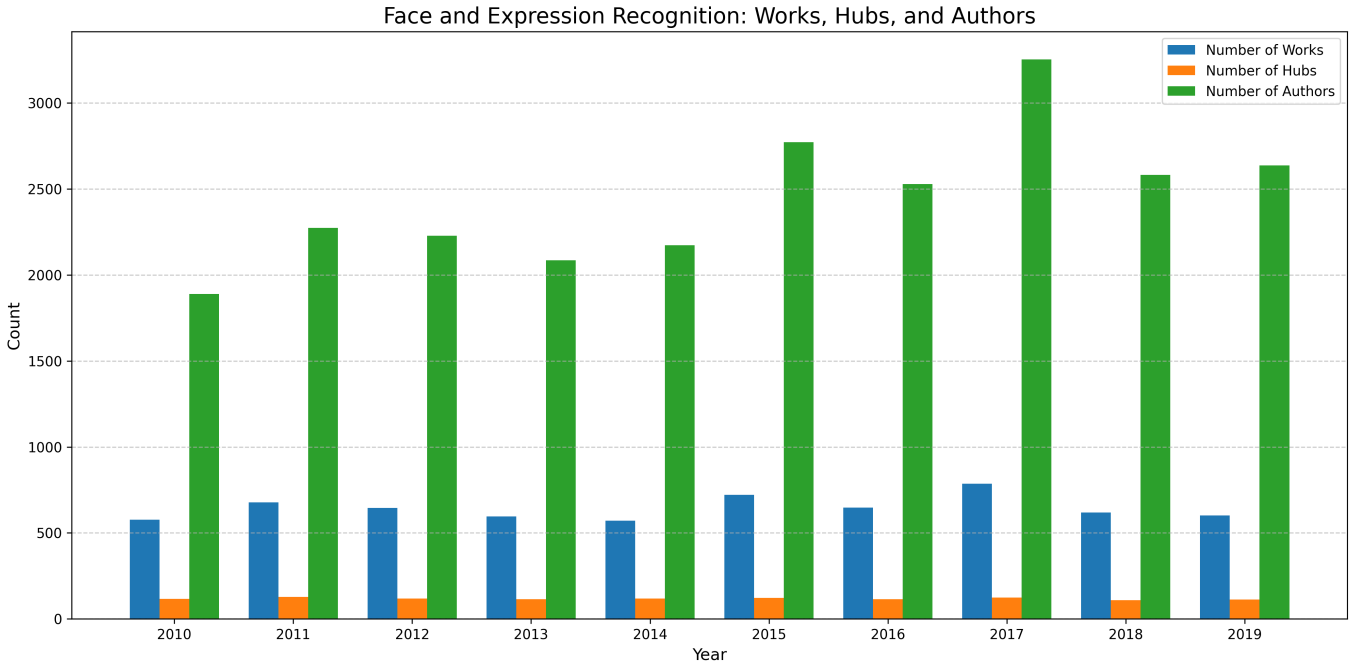
Face and Expression Recognition: Works, Hubs, and Authors
Number of Works
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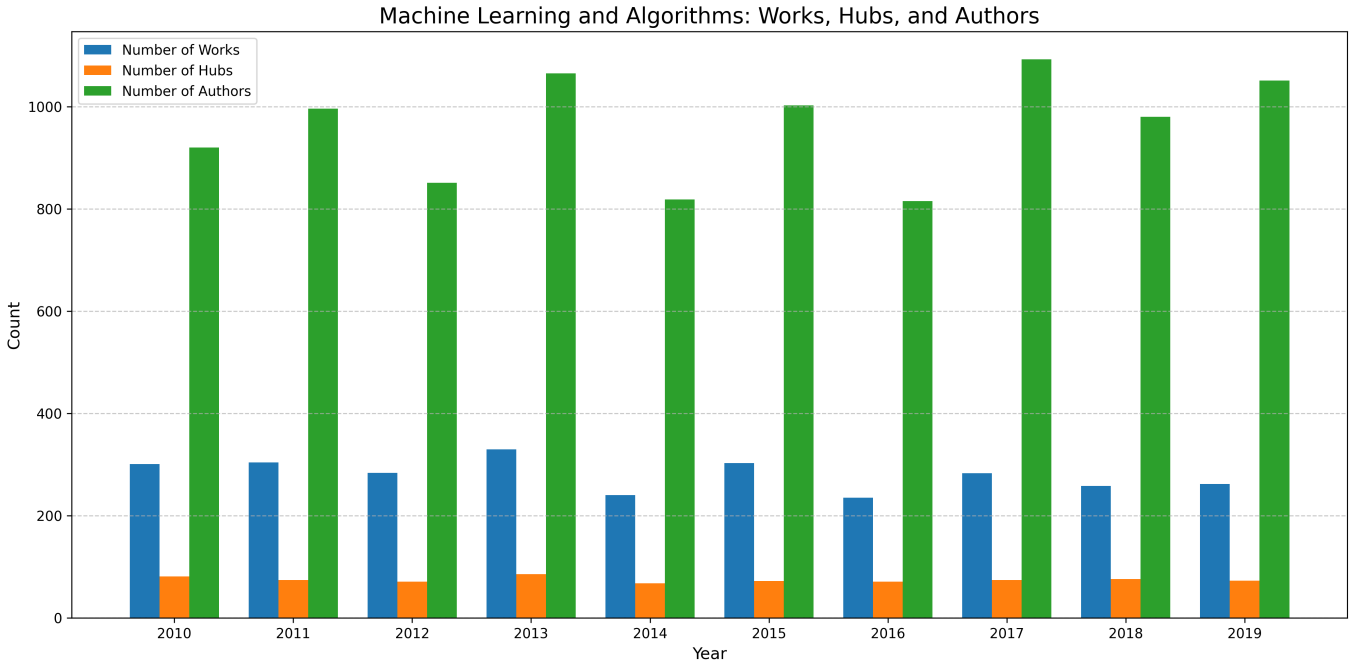
Fig. 16.


Machine Learning and Algorithms: Works, Hubs, and Authors
Number of Works
Number of Hubs
Number of Authors
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